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$$f(x) = \tan x - \frac{1 - \cos x}{\sin x}$$

$$f'(x) = \frac{d}{dx}(\tan x) - \frac{d}{dx} \left(\frac{1 - \cos x}{\sin x} \right)$$

$$\frac{d}{dx}(\tan x) = \sec^2 x$$

$$\frac{d}{dx} \left(\frac{1 - \cos x}{\sin x} \right) = \frac{\sin x \cdot \sin x - (1 - \cos x) \cdot \cos x}{\sin^2 x}$$

$$= \frac{\sin^2 x - \cos x + \cos^2 x}{\sin^2 x}$$

$$= 1 - \frac{\cos x \cdot (1 - \cos x)}{\sin^2 x}$$

$$f'(x) = \sec^2 x - \left(1 - \frac{\cos x \cdot (1 - \cos x)}{\sin^2 x} \right)$$

References